

A cardinal Jung dichotomy for spaces $C(K)$

Agent lane 11

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Source and Target

The source paper is J. M. F. Castillo and P. L. Papini, *The separable Jung constant in Banach spaces*, arXiv:1910.02402. In the subsection “On the stability of the Jung constants”, the authors ask whether, for a space $C(K)$ of continuous functions on a compact space, the inequality $J_{\aleph}(C(K)) < 2$ implies $J_{\aleph}(C(K)) = 1$.

This packet proves that question for every uncountable cardinal $\aleph$. The exact result is the dichotomy

$$J_{\aleph}(C(K)) \in \{1, 2\},$$

with the value determined by whether K is an $F_{\aleph}$ -space.

Definitions

For a bounded set A in a Banach space X , write

$$\delta(A) = \sup\{\|a - b\| : a, b \in A\}, \quad r(A) = \inf_{x \in X} \sup_{a \in A} \|a - x\|.$$

For an uncountable cardinal $\aleph$, the cardinal Jung constant is

$$J_{\aleph}(X) = \sup \frac{2r(A)}{\delta(A)},$$

where the supremum is over closed bounded sets $A \subset X$ with $\delta(A) > 0$ and $\text{dens}(A) < \aleph$.

A compact Hausdorff space K is an $F_{\aleph}$ -space if every two disjoint open subsets of K which are unions of fewer than $\aleph$ closed sets have disjoint closures.

Proof Intuition

The positive direction is already encoded in the cited injectivity theory: $J_{\aleph}(X) = 1$ means $(1, \aleph)$ -injectivity, and for $C(K)$ spaces that is exactly the $F_{\aleph}$ property of K .

For the converse, the failure of $F_{\aleph}$ supplies two disjoint open sets G, H , built from fewer than $\aleph$ closed pieces, whose closures touch. Continuous functions supported on the closed pieces of G demand that any small-radius center be positive on $\overline{G}$; the negative of the analogous functions on H demands that it be negative on $\overline{H}$. At a common closure point these demands are incompatible.

Lemma 1. *Let $\aleph$ be an uncountable cardinal. If K is not an $F_{\aleph}$ -space, then $C(K)$ contains a closed bounded set A with $\text{dens}(A) < \aleph$, $\delta(A) = 1$, and $r(A) = 1$.*

Proof. Since K is not an $F_{\aleph}$ -space, there are disjoint open sets

$$G = \bigcup_{i \in I} C_i, \quad H = \bigcup_{j \in J} D_j$$

with $|I|, |J| < \aleph$, where each C_i, D_j is closed in K , and

$$\overline{G} \cap \overline{H} \neq \emptyset.$$

For each $i \in I$, Urysohn's lemma gives $u_i \in C(K)$ with

$$0 \leq u_i \leq 1, \quad u_i|_{C_i} = 1, \quad u_i|_{K \setminus G} = 0.$$

For each $j \in J$, choose $v_j \in C(K)$ with

$$0 \leq v_j \leq 1, \quad v_j|_{D_j} = 1, \quad v_j|_{K \setminus H} = 0.$$

Set

$$A_0 = \{u_i : i \in I\} \cup \{-v_j : j \in J\} \subset C(K),$$

and let $A = \overline{A_0}$ in $C(K)$. Since $|I| + |J| < \aleph$, the set A_0 has cardinal $< \aleph$, and therefore $\text{dens}(A) < \aleph$.

All u_i 's and v_j 's take values in $[0, 1]$, and the supports of the u_i 's are disjoint from the supports of the v_j 's. Hence

$$\|u_i - u_{i'}\| \leq 1, \quad \|-v_j + v_{j'}\| \leq 1, \quad \|u_i - (-v_j)\| = \|u_i + v_j\| \leq 1.$$

Thus $\delta(A) \leq 1$. Since G and H are nonempty, choose $i \in I$ with $C_i \neq \emptyset$ and $j \in J$. At any $x \in C_i$ we have $u_i(x) = 1$ and $v_j(x) = 0$, so $\|u_i - (-v_j)\| \geq 1$. Therefore $\delta(A) = 1$.

The zero function is a center of radius 1, so $r(A) \leq 1$. Suppose, toward a contradiction, that $h \in C(K)$ and $\rho < 1$ satisfy

$$\sup_{a \in A} \|h - a\| \leq \rho.$$

For every $i \in I$ and every $x \in C_i$,

$$\text{Re } h(x) \geq 1 - \rho,$$

because $u_i(x) = 1$. Hence $\text{Re } h \geq 1 - \rho$ on G , and by continuity also on $\overline{G}$. Similarly, for every $j \in J$ and every $y \in D_j$,

$$\text{Re } h(y) \leq -1 + \rho,$$

because $-v_j(y) = -1$. Hence $\text{Re } h \leq -1 + \rho$ on $\overline{H}$. At any point $p \in \overline{G} \cap \overline{H}$, this gives

$$1 - \rho \leq \text{Re } h(p) \leq -1 + \rho,$$

which is impossible for $\rho < 1$. Thus $r(A) \geq 1$, and so $r(A) = 1$. □

Theorem 1. *Let $\aleph$ be an uncountable cardinal. For every compact Hausdorff space K ,*

$$J_{\aleph}(C(K)) = \begin{cases} 1, & K \text{ is an } F_{\aleph}\text{-space,} \\ 2, & K \text{ is not an } F_{\aleph}\text{-space.} \end{cases}$$

Proof. If K is an $F_{\aleph}$ -space, then the characterization of Aviles–Cabello Sanchez–Castillo–Gonzalez–Moreno says that $C(K)$ is $(1, \aleph)$ -injective. Castillo–Papini prove that a Banach space X is $(1, \aleph)$ -injective if and only if $J_{\aleph}(X) = 1$. Hence $J_{\aleph}(C(K)) = 1$.

If K is not an $F_{\aleph}$ -space, Lemma 1 gives a closed bounded set $A \subset C(K)$ with $\text{dens}(A) < \aleph$, $\delta(A) = 1$, and $r(A) = 1$. Therefore

$$J_{\aleph}(C(K)) \geq \frac{2r(A)}{\delta(A)} = 2.$$

For every Banach space X , $J_{\aleph}(X) \leq 2$, since every bounded set has radius at most its diameter by centering at one of its own points. Consequently $J_{\aleph}(C(K)) = 2$. $\square$

Corollary 1. *For every uncountable cardinal $\aleph$ and compact Hausdorff space K , if $J_{\aleph}(C(K)) < 2$, then $J_{\aleph}(C(K)) = 1$.*

Verification and Novelty Check

The cheap run indexes were searched for the arXiv id, the paper title, and the phrases $J_s(C(K))$, $J_{\aleph}(C(K))$, separable Jung constant, Jung constant $C(K)$, F -space, and $F_{\aleph}$ -space. The only direct hit for this target was this lane’s earlier separable packet, now upgraded in place.

The local source of arXiv:1406.6733 was checked for the definition of $F_{\aleph}$ -spaces and the characterization of $(1, \aleph)$ -injective $C(K)$ spaces. Web searches on 2026-07-03 for exact phrases including “ $J_{\aleph}(C(K))$ ”, “ $F_{\aleph}$ Jung constant $C(K)$ ”, and “The separable Jung constant in Banach spaces $J_{\aleph}$ ” found the source paper and the cited cardinal-injectivity paper, but no close exact resolution.

References

- J. M. F. Castillo and P. L. Papini, *The separable Jung constant in Banach spaces*, arXiv:1910.02402.
- A. Aviles, F. Cabello Sanchez, J. M. F. Castillo, M. Gonzalez, and Y. Moreno, $\aleph$ -*injective Banach spaces and $\aleph$ -projective compacta*, Rev. Mat. Iberoam. 31 (2015), 575–600; arXiv:1406.6733.